Primitive DATA TYPES

A data type tells Python what kind of value a variable holds, like a number, text, or a list.

Primitive data type

Non- Primitive data type

Primitive data type

int

float

string

boolean

complex

Non Primitive data type

List

Tuple

Set

dictonary

Primitive Data type

A primitive data type is a basic, built-in data type that stores a single, simple value and cannot be broken down into smaller parts.

It is immutable, which means its value cannot be changed after it is created.

Numeric type

1.integer

Creating whole numbers.Positive, negative, and zero

In [61]:
```
num = int(input("enter num:"))
print(num)
print(type(num))
```

```
-19
<class 'int'>
```

2.float

Creating decimal numbers

In [37]:
```
num1 = float(input("enter num1:"))
print(num1)
print(type(num1))
```

```
23.5
<class 'float'>
```

3.Complex

Real and imaginary part.real and .imag

In [40]:
```
num2 = complex(input("enter num2:"))
print(num2)
print(type(num2))
```

```
(3+7j)
<class 'complex'>
```

Text type

4.String

Creating strings (single, double, triple quotes)

In [44]:
```
Name = input("enter name:")
print(Name)
print(type(Name))
```

```
Reeta
<class 'str'>
```

Boolean type

5.boolean

True and False.Truthy and falsy values

In [49]:
```
grade = bool(input("enter grade:"))
print(grade)
print(type(grade))
```

```
True
<class 'bool'>
```

In [51]:
```
grade = bool(input("enter grade:"))
print(grade)
print(type(grade))
```

```
True
<class 'bool'>
```

In [53]:
```
grade = bool(input("enter grade:"))
print(grade)
print(type(grade))
```

```
True
<class 'bool'>
```

In [55]:
```
grade = bool(input("enter grade:"))
print(grade)
print(type(grade))
```

```
True
<class 'bool'>
```

In [57]:
```
grade = bool(input("enter grade:"))
print(grade)
print(type(grade))
```

```
True
<class 'bool'>
```

In [59]:
```
grade = bool(input("enter grade:"))
print(grade)
print(type(grade))
```

```
True
<class 'bool'>
```

None type

NoneType is the data type of the value None. None means "no value" or "nothing".

In [74]:
```
a=None
print(a)
print(type(a))
```

```
None
<class 'NoneType'>
```

Practice Questions

1. Write a program to print the data type of these variables: a = 10, b = 3.14, c = 3 + 4j, d = "Hello", e = True, f = None.

In [85]:
```
a=10
print(type(a))
```

```
<class 'int'>
```

In [87]:
```
b=3.14
print(type(b))
```

```
<class 'float'>
```

In [89]:
```
c=3+4j
print(type(c))
```

```
<class 'complex'>
```

In [91]:
```
d="hello"
print(type(d))
```

```
<class 'str'>
```

In [93]:
```
e=True
print(type(e))
```

```
<class 'bool'>
```

In [95]:
```
f=None
print(type(f))
```

```
<class 'NoneType'>
```

2. Write a program to take two numbers from the user and print their sum, difference, product, and division.

In [98]:
```
num1 = int(input("enter num1:"))
num2 = int(input("enter num2:"))
sum = num1 + num2
print("sum:",sum)
diff = num1 - num2
print("difference:",diff)
pro = num1 * num2
print("product:",pro)
div = num1 / num2
print("division:",div)
```

```
sum: 13
difference: 7
product: 30
division: 3.3333333333333335
```

3. Write a program to print the real and imaginary parts of the complex number z = 5 + 8j.

In [101]:
```
com = 5+8j
print("real:",com.real)
print("imaginary:",com.imag)
```

```
real: 5.0
imaginary: 8.0
```

4.Write a program to convert the string "100" into an integer and a float, then print both with their types.

In [108]:
```
num1=input("enter num1:")
print(type(num1))
num2=int(num1)
print("num2:",num2)
print(type(num2))
num3 = float(num1)
print("num3:",num3)
print(type(num3))
```

```
<class 'str'>
num2: 100
<class 'int'>
num3: 100.0
<class 'float'>
```

5.Write a program to convert the float 7.9 into an integer and print the result and its type.

In [118]:
```
num1 = float(input("ener num1:"))
print(type(num1))
num2 = int(num1)
print(num2)
print(type(num2))
```

```
<class 'float'>
7
<class 'int'>
```

6.Write a program to show that 0.1 + 0.2 is not equal to 0.3, and fix it using round()

In [123]:
```
num1 = 0.1 + 0.2
num2 = 0.3
n =(round(num1,2)==num2)
print(n)
```

```
True
```

7.Write a program to show that a string is immutable by trying to change its first character.

In [128]:
```
a = "Hello World"
a[0] = "h"
print(a)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[128], line 2
      1 a = "Hello World"
----> 2 a[0] = "h"
      3 print(a)

TypeError: 'str' object does not support item assignment
```

8.Write a program to check the type of the result when you add int and float (5 + 2.5).

In [131]:
```
num1 = 5
num2 = 2.5
num3 = num1 + num2
print(num3)
print(type(num3))
```

```
7.5
<class 'float'>
```

9.Write a program to print the bool value of these: 0, 1, "", "Hi", None, 0.0, -5

In [135]:
```
a=0
print(bool(a))
b=1
print(bool(b))
c="hi"
print(bool(c))
d=None
print(bool(d))
e=0.0
print(bool(e))
f=-5
print(bool(f))
```

```
False
True
True
False
False
True
```

10.Write a program to show that True + True gives 2 and print its type

In [138...

```
a=True
b=True
print(a+b)
```

```
2
```